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Reflection Journal

From knowing barely anything about coding, I think the most I have learned is through the final. I understand how important it is to be aware of the libraries. I was able to work on this and comprehend why variables or functions worked the way they did. As a group it was nice working, but it would've been better if I understood better how to code. Even though I was able to complete this without my teammates, I can say that I feel a bit more confident with training models. At first, when I read that we had to train and evaluate six models, I was very intimidated. I kept thinking to myself, "How will I be able to do this if I barely even know a bit of coding?" Fortunately, I pushed through, I used resources, tools, whatever I could put my hands on to have a functioning code. Using my group's midterm as a reference, I reorganized and changed variables to be readable. For someone with zero to a bit of experience, if they were to read the previous code, it could seem a bit scattered. Back to the six models, I did not think it would be easy to code, I genuinely thought that I would have to spend six hours writing each model's code. However, in my intro to programming course, we talked about loops. I did not want to waste time with each individual model, that would be tedious. The variables that the models required were different to some but the same for others and really the only thing that needed to be changed was the name of the model in parts of the code. However, I slipped up by having the results variable within the loop. I was stuck on why the results would stack up and not simply give me the individual results of the model. The solution for that was easy because it only required me to get out of the loop. One of the parts that was quite time consuming was creating the voting classifier and the average ensemble models. That is because lots of variables had to be changed so that the program wouldn't get confused on the repetition of variables. However, the evaluation of the models using the metrics accuracy, precision, recall, f1 score, and roc-auc score, was repetitive and only required me to change the variables where needed. Same for the average ensemble model, the only thing that had a different approach was how the models would be stored. I was surprised to see that once compared, both displayed the same results (in the validation and test dataset) and that's how I got to learn that it's not necessarily a bad thing but maybe training with different or more diverse data could let me see which model performed better in respect to the metrics used to evaluate the models. The Bayesian model was also pretty similar to the other ensembles in terms of organization and how to write it! Which was of relief to me. In this lab, I was able to appreciate the concept of "wisdom of the crowd". Individually, some models did carried out positive results by themselves and only had one or two metrics in which they lacked a better performance, but by uniting them, the results lean towards more confident results.